

Suppose that this preferred portfolio has expected return $u^* = 10\%$ and risk $\sigma^* = 20\%$. The trader then performs trade schedule optimization (Equation 10.4) and constructs the efficient trading frontier (ETF) using values of $0 \leq \lambda \leq 10$ to determine the best way to implement the portfolio.

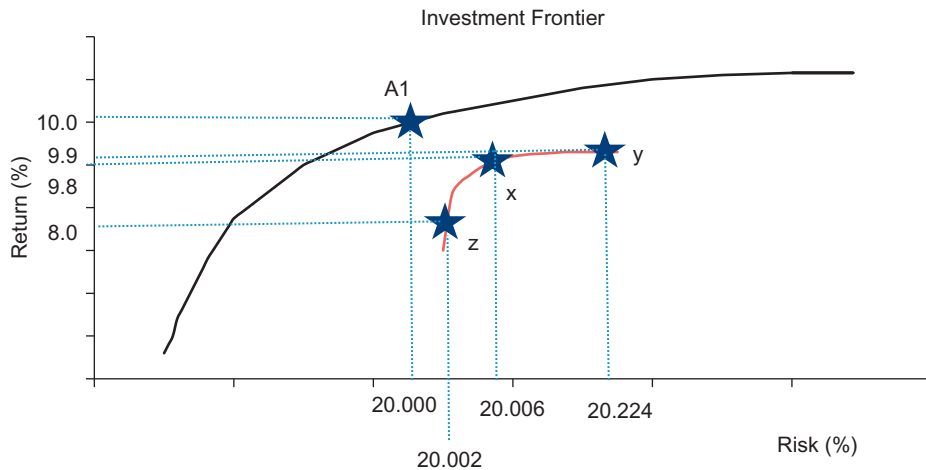
Rather than analyze our trade schedules in the traditional cost-risk space, let us examine our trading cost consequences by overlaying the efficient trading frontier on the efficient investment frontier (Figure 10.4). Notice that the efficient trading frontier is now inverted from its more traditional appearance and shows the cost-adjusted potential risk-return profile for optimal portfolio A1. The efficient portfolio A¹ is no longer associated with a single expected return and risk. There are multiple sets of potential return and risk depending upon the trading strategy.

The adjusted return for the portfolio will be reduced by the estimated impact cost. That is:

$$\text{Adjusted Return} = \text{Portfolio Return} - \text{Strategy (Cost)} \quad (10.5)$$

The new timing risk, however, will increase due to the market exposure incurred while acquiring the position. The actual increase in risk is additive in variance (risk value squared). We add the one day timing risk to the annualized portfolio risk value. This is calculated as follows:

$$\text{Adjusted Risk} = \sqrt{(\text{Portfolio Risk})^2 + (\text{Strategy Risk})^2} \quad (10.6)$$



■ Figure 10.4 Investment Frontier with Trading Frontier as an Overlay. Source: *Journal of Trading* (2007).